

Nandha R

Data Scientist & AI / ML Engineer

Chennai (Current) & Dharmapuri (Native), Tamil Nadu

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Objective: To work in an environment offering challenging career opportunities where I can apply and enhance my skills for solving diverse technical problems across Data Science, Machine Learning, and AI Systems.

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL

AI & Machine Learning: AutoML, XGBoost, LightGBM, CatBoost, Scikit-learn, RAG, ChromaDB, Ollama, Predictive Modeling, Regression, Classification, Feature Engineering, Hyperparameter Tuning, Benchmarking, EDA

NLP & Speech AI: Speech Recognition (SR), Natural Language Processing (NLP), Text-to-Speech (TTS), Sentence Transformers, BM25 Hybrid Search

Frameworks & Tools: FastAPI, React.js, Streamlit, Docker, Python Data Pipelines

EXPERIENCE

Data Scientist Intern — Gradtwain (Chennai, Tamil Nadu) June 2026 – August 2026

- Architected end-to-end Python data pipelines and performed rigorous EDA on multi-source datasets to uncover actionable business insights for client use cases.
- Developed and optimized predictive modeling workflows—including regression and classification pipelines—improving forecasting accuracy through systematic feature engineering and model benchmarking.
- Designed and deployed a production-grade RAG AI chatbot, integrating LLM APIs, ChromaDB vector storage, and semantic document retrieval to enhance operational workflows.
- Cleaned, transformed, and engineered features from complex, high-volume datasets, accelerating data preprocessing pipelines.
- Collaborated cross-functionally with engineering and product teams to implement ML-driven solutions that directly improved operational efficiency.

FEATURED PROJECTS

AI-Powered Data Cleaning and Imputation Engine (DataClean AI) Aug 2026

Stack: FastAPI, React.js, Scikit-learn, XGBoost, LightGBM, CatBoost, Docker.

- Automated Quality Detection:** Autonomously detects 13+ data anomalies including missingness patterns (MCAR/MAR), multivariate outliers (IQR/Z-score/MAD), distribution skewness, and class imbalance.
- Hybrid AI Recommendation:** Combines 30+ domain heuristic rules with a Voting Ensemble classifier (XGBoost + LightGBM + CatBoost) achieving 97.8% recommendation accuracy across 25 cleaning techniques.
- Scalable Streaming Engine:** Implements a 10MB chunked multipart upload pipeline handling multi-gigabyte datasets (3M+ rows) with zero-OOM memory streaming and automated PDF audit reporting.

PDF Summarizer Chatbot using RAG (PaperBrain) Jul 2026

Stack: FastAPI, Streamlit, ChromaDB, Sentence Transformers, BM25, Ollama (Local LLMs).

- Local & Privacy-First:** Fully offline RAG system where embeddings, vector storage, and language generation run locally without data leaving the machine.
- Hybrid Search:** Fuses dense vector similarity search (Sentence Transformers) with BM25 keyword matching for context retrieval.

The Voice Assistant with Desktop Accessibility Mar 2026

Stack: Python, Speech Recognition (SR), NLP Intent Engine, Text-to-Speech (TTS), OS Automation.

- Hands-free desktop voice assistant with Speech Recognition, NLP command parsing, and responsive TTS audio feedback for desktop navigation.

Residential Property Valuation Using Machine Learning Feb 2025

Stack: Python, Scikit-Learn, Pandas, NumPy, Matplotlib, Seaborn.

- Engineered regression modeling workflows on large real estate datasets to accurately forecast property market valuations based on key features.

EDUCATION

BE – Computer Science and Engineering (Vinayaka Mission's KVEC, Salem • 2022–2026)	73.76%
Higher Secondary School Examination (HSC) (Govt Higher Secondary School, Salem • 2021–2022)	52.8%
Secondary School Examination (SSLC) (Govt Higher Secondary School, Salem • 2019–2020)	72.5%

CERTIFICATES & ACHIEVEMENTS

- GenAI Powered Data Analytics Job Simulation (Jan 2026)** — Completed Tata (via Forage) simulation in EDA, AI risk profiling, and delinquency prediction. [Verification: [SFPHHSaWj4nHTiKK7](#)]
- 1st Prize Winner – Kongu Engineering College IPL AUCTION 2025** — Analytics, valuation, and bidding strategy optimization.
- Data Science in Python Training Program (2024)** — IIIT KOTTAYAM Engineering College intensive training program.
- Green Skill and AI Course Certification (2025)** — Edunet Foundation course on AI fundamentals & sustainability.
- Paper Presentation on 'IOT in Military Security' (2023)** — KSR Engineering College technical presentation.